

Predicting Protein Secondary Structure by a Support Vector Machine Based on a New Coding Scheme

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Abstract

Protein structure prediction is one of the most important problems in modern computational biology. Protein secondary structure prediction is a key step in prediction of protein tertiary structure. There have emerged many methods based on machine learning techniques, such as neural networks (NN) and support vector machine (SVM) etc., to focus on the prediction of the secondary structures. In this paper, a new method was proposed based on SVM. Different from the existing methods, this method takes into account of the physical-chemical properties and structure properties of amino acids. When tested on the most popular dataset CB513, it achieved a Q_3 accuracy of 0.7844, which illustrates that it is one of the top range methods for protein of secondary structure prediction.

Keywords: protein secondary structure prediction, SVM, coding scheme, CB513

1 Introduction

The structure of a protein reveals important information, such as the location of probable functional or interaction sites, identification of distantly related proteins, and discovery of important regions of the protein sequence that are involved in maintaining the structure, and so on. Along with a large number of protein sequences having been produced in high-throughput experiments, prediction of protein structure and function from amino acid sequences becomes one of the most important problems in current time, and many researches have focused on the methods of protein structure determination. It is very difficult and time expensive to experimentally determine protein structures (such as X-ray crystallography and Nuclear Magnetic Resonance spectroscopy techniques) [21], therefore computational methods to predict structures have been rigorously explored. The study of protein secondary structure plays an important role in protein tertiary structure prediction with the *ab initio* method [2] and attracts more attentions.

Since 1970s, many approaches for predicting secondary structures from sequences have been developed [20]. In order to improve the prediction accuracy, two kinds of efforts have been done: one is the design of amino acid coding scheme. Early coding scheme only used 21-dimension arrays to denote each amino acid without any biological meanings [12, 22]. And later, coding scheme contained more evolutionary information. For examples, the PHD program developed by Rost and Sander [25, 26] used multiple sequences alignments for the first time; the PSIPRED program stated by Jones [15] used position-specific scoring matrices obtained in PSI-BLAST searches [1]. The accuracies of these methods achieve Q_3 scores between 70% and 80%, where Q_3 is the percentage of amino acids correctly predicted as helix, sheet, or coil if all amino acids are classified in one of the three groups; The other effort is the application of machine learning methods, such as information theory [9], neural

network [12, 22] and support vector machine [10, 14, 17]. Q_3 scores produced by information theory method ranked between 57%~66%; simple neural network with the simple coding scheme achieved Q_3 scores about 63%~65%; and support vector machine can usually get 74%~80% Q_3 accuracies [9, 10, 12, 14, 17, 22]. It should be noted that, these methods were performed on different data sets. Early methods were experimented on small data sets, and didn't consider the homologous relationship among them. In recent years, two widely data sets RS126 [26] and CB513 [5] are used to evaluate the performance of different methods. Sequences in RS126 and CB513 are non-homologous. The top range overall per-residue accuracy Q_3 on CB513 varied from 73.5%~76.6% [5, 10, 14, 17]. In this paper, we will test our method on this data set to get the Q_3 accuracies.

SVM is a supervised machine learning technology and well founded theoretically on statistical learning theory [7, 11, 13, 24, 27], so we still use the support vector machine to predict protein secondary structures in this paper. Different from previous methods, new coding schemes based on the physical-chemical properties and the structure properties of proteins are adopted. The rest of this paper is organized as follows: In Section 2, we present the materials used in the following experiments. In Section 3, we describe our method in details, including the coding scheme, finding out the optimal window length and building tertiary classifiers, etc. Section 4 is the evaluation results of our method on CB513 data set, and the best Q_3 accuracy is 78.44%. Finally, we end with the conclusions in the last section.

2 Materials

Even for a protein of known structure, the assignment of a secondary state to each residue is not a straightforward task [18]. Three programs are currently widely used to define the secondary structure of a protein from its atomic coordinates: DSSP [16], STRIDE [8] and DEFINE [23]. We use DSSP because it's the most widely used software. It has eight secondary structure classes: H(α -helix), G(3_{10} -helix), I(π -helix), E(β -strand), B(isolated β -bridge), T(turn), S(bend) and -(rest). They are usually reduced to three states by one of the following methods:

- (1) H, G and I to **H**; E to **E**; the rest to **C**
- (2) H, G to **H**; E, B to **E**; the rest to **C**
- (3) H, G to **H**; E to **E**; the rest to **C**
- (4) H to **H**; E, B to **E**; the rest to **C**
- (5) H to **H**; E to **E**; the rest to **C**

Although method (5) obviously increases the accuracy [5], and method (2) usually leads a reduction of prediction accuracy, in order to provide fair comparison of our results with other methods, we still use method (2).

To compare our new method with some previously published methods [14, 17], we selected the non-homologous RS126 and CB513 data sets. The information of RS126 was obtained from [26] and their secondary structures were downloaded from the RCSB Protein Data Bank: <http://www.rcsb.org/pdb/>. This set contains 23,411 residues, and has a composition of 31.16% helix, 23.20% sheet, and 45.63% coil. The CB513 was obtained from this website: <http://www.compbio.dundee.ac.uk/~www-jpred/>. It contains 84,034 residues, and has a composition of 34.30% helix, 22.54% sheet, and 43.16% coil.

All the protein sequences represent unique protein folds. To get more reliable results, two datasets have experienced the 7-fold cross-validation test. Each dataset was divided into seven folds with a similar number of proteins and similar composition of the secondary structures. In each time, one fold was selected as the testing set and the others were used as the training set. The whole procedure was irritated for seven times. We used RS126 to find out the optimal window length for prediction. The prediction accuracy of CB513 was presented to compare with other methods.

3 Methods

3.1 The Support Vector Machine

The support vector machine (SVM) constructs an optimal separating hyperplane, which maximizes the margin (i.e. the distance between the hyperplane and the nearest data point of each class) by mapping the input space into a high-dimensional feature space. The mapping is determined by a kernel function. Many research results show that SVMs has crucial advantages such as fast convergence (typically about 1-2 orders of magnitude faster than neural networks (NNs) [6]), tending not to over-hit and the ability to find the problem formulation as a quadratic convex function minimization that is easier to solve [14], and so on.

In this paper, we used a L_1 soft margin support vector machine [30] to build the binary classifier, and chose the RBF (radial basis function) kernel defined as:

$$K(\vec{x}_i, \vec{x}_j) = \exp\left(-\gamma|\vec{x}_i - \vec{x}_j|^2\right) \quad (1)$$

When constructing the SVM classifiers, we set parameter γ to 0.10 and the regularization parameter C to 1.50. These parameters were chosen by experience.

All the programs are written in MATLAB. We downloaded osu_svm classifier Matlab toolbox (Version 3.00) from http://www.ece.osu.edu/~maj/osu_svm/osu_svm3.00.zip. By virtue of Matlab's mex mechanism, this toolbox implements SVM classifiers in C++ using Chih-Chung Chang and Chih-Jen Lin's LIBSVM algorithm (COPYRIGHT_libsvm) [3], therefore it is comparable to all the available C-coded SVM packages in speed. Meanwhile, it inherits the memory-management mechanism of LIBSVM, and is capable of dealing with practical classification problem with huge training sets.

3.2 Prediction Accuracy Assessments

Several standard performance measures can be used to assess prediction accuracy. Here we introduce two of them shown as formula (2) and (3). Q_3 is a measure of the three-state overall percentage of correctly predicted residues:

$$Q_3 = \frac{P_\alpha + P_\beta + P_{coil}}{N} \quad (2)$$

Where N is the total number of predicted residues and P_α is the number of correctly predicted secondary structures of type α (α -helix). The accuracy for type α can be calculated as:

$$Q_\alpha = \frac{P_\alpha}{\text{total number of residue in } H} \quad (3)$$

Q_α and Q_{coil} can be defined similar to Q_α .

3.3 Coding Schemes for Vector Space to Represent Protein

As the neural network approach, we adopted the classical local coding scheme of the protein sequence with a sliding window [12]. The SVM encodes a moving window in the amino acid sequence and prediction is made for the central residue in the window. A vector was used to encode each amino acid residue, in which, one or several consecutive elements represent a kind of physical or chemical characteristics and properties of amino acids. In this paper, we present three coding schemes to discuss the different effects of different characters to the structure of proteins.

A Coding Scheme 1

Each amino acid has at least one amine and one acid functional group as the name implies. The different properties result from variations in the structures of different R groups. The R group is often referred as the amino acid side chain.

Side chains that have various functional groups such as acids, amides, alcohols, and amines will impart a more polar character to the amino acid. Figure 1 illustrates several properties of amino acids.

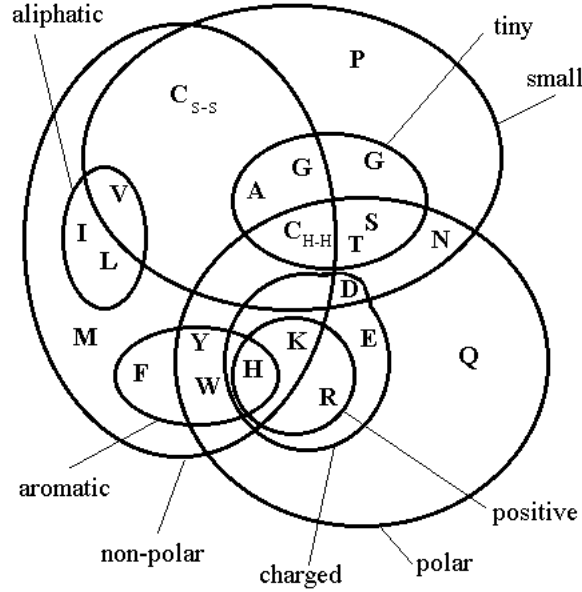


Figure 1: A Venn diagram showing the relationship of the 20 naturally occurring amino acids that is useful for the selection of physical-chemical properties thought to be important in the determination of protein structure.

In the process of protein folding, polar residues prefer to stay outside of the protein to prevent the non-polar residues from exposing to the polar solvent molecular, such as water molecular. The interactions between the non-polar residue side chains are called to be hydrophobic interactions.

Coding scheme 1 makes use of the influence of hydrophobic interactions to the secondary structure. Here, a single number codes each residue in the protein sequence, and this number indicates the hydrophobic degree of the corresponding residue. There are many methods to display the parameter of hydrophobic characters of the 20 amino acid residues. Because different labs present the parameters based on different foundations, the parameters vary a lot. But the relative relationship of their hydrophobic characters is consistent with each other. We use the parameters obtained by Kyte and Doolittle's method [29], here denoted as K-D method, which is one of the most widely used methods. They summarized the distributing coefficients of amino acids in organic solvents and water and the distribution of amino acids in protein structures.

The hydrophobic character parameters of the 20 amino acid residues obtained by K-D method are: I, 4.5; V, 4.2; L, 3.8; F, 2.8; C, 2.5; M, 1.9; A, 1.8; G, 0.4; T, -0.7; S, -0.8; W, -0.9; Y, -1.3; P, -1.6; H, -3.2; E, -3.5; N, -3.5; Q, -3.5; N, -3.5; K, -3.9; R, 4.0.

B Coding Scheme 2

In the second coding scheme, we use a 3-dimension vector to code an amino acid. In this vector, the first unit stands for the conformation propensity factor of α -helix for the corresponding amino acid residue, the second is for β -sheet, and the last one is for coil. The conformation parameters for each amino acid P_{ij} are defined as follow:

$$P_{ij} = f_{ij}/f_j, \quad i = 1, \dots, 20, \quad j = 1, \dots, 3 \quad (4)$$

Where f_j is the score of all of the amino acids residues whose secondary structure is the j th conformation of the 3 types; f_{ij} is the corresponding score of the i th amino acid. They can be calculated respectively by the following formula:

$$f_{ij} = n_{ij}/N_i \quad (5)$$

$$f_j = N_j/N_t \quad (6)$$

In these formula, j indicates the 3 types secondary conformation: H, E and C; i indicates the 20 amino acid: A, R, N, D, C, Q, E, G, H, I, L, K, M, F, P, S, T, W, Y, V. In the statistical sample, n_{ij} is the total number of i th residues that has j th conformation; N_t is the total number of residues,

$$N_t = \sum_{i=1}^{20} \sum_{j=1}^3 n_{ij}; N_i \text{ is the total number of the } i\text{th amino acid residues, } N_i = \sum_{j=1}^3 n_{ij}; N_j \text{ is the total}$$

number of the j th conformation, $N_j = \sum_{i=1}^{20} n_{ij}$.

Obviously, $P_{ij} > 1$ indicates that the i th residue has the propensity to form j th conformation, and $P_{ij} < 1$ indicates that it is likely to form other kinds of conformation. If $P_{i\alpha} > 1$, we call the word i is a helix former; otherwise, we call it a helix breaker. β -Sheet former has a familiar definition. In the experiments, all the conformation parameters P_{ij} are calculated from the training set, and changed as the training sets become different.

C Coding Scheme 3

We want to know whether the prediction accuracy would increase when more characters are considered. So in coding scheme 3, we combine scheme 1 and 2 and use a vector of 4-dimension to code the amino acid. The first three elements are the same as scheme 2; the last one is the same as scheme 1. Table 1 illustrates one such scheme from a training set. The first three elements of the coding vector for each amino acid were calculated from the training set, and the last one was from K-D method, which were fixed during the whole experiment procedure, and the higher values indicate the amino acid are with more stronger hydrophobicity.

3.4 Searching the Optimal Window Length For Binary Classifiers

The optimal window length of the sliding window-coding scheme was obtained by investigating the accuracy of various window sizes on different coding schemes for amino acids. When the window size is too small, it may lose some important classification information and lead to low prediction accuracy, and a too large window size may suffer from inclusion of unnecessary noise.

To find the optimal window length, we carry on three experiments on RS126 for the three coding schemes, we calculate the prediction accuracy of the six binary classifiers, H/ $\sim$ H, E/ $\sim$ E, C/ $\sim$ C, H/E, H/C, E/C, for $L = 3, 5, \dots, 17, 19, 21$. Results in Table 2, Table 3 and Table 4 show that the prediction accuracy for the binary classifier does not change dramatically when a window length is larger than 15. This is slightly smaller than the sliding window length of 17 of the first layer neural network for sequence to structure prediction in Jnet [4], which shows that SVM method can effectively deal with noise. We choose a window length of 15 for all the results of this paper.

Table 1: Coding scheme for each amino acid.

	<i>H</i>	<i>E</i>	<i>C</i>	<i>K-D</i>
A	1.4587	0.6962	0.8491	1.8
R	1.2498	0.8031	0.9393	-4.0
N	0.8040	0.6428	1.3278	-3.5
D	0.9671	0.5056	1.2845	-3.5
C	0.6698	1.4103	1.0328	2.5
Q	1.2532	0.7660	0.9557	-3.5
E	1.4630	0.6267	0.8812	-3.5
G	0.4867	0.6879	1.5240	0.4
H	0.9192	1.1220	1.0063	-3.2
I	1.0150	1.6853	0.6558	4.5
L	1.3375	1.1946	0.6811	3.8
K	1.1249	0.8516	1.0009	-3.9
M	1.3512	1.1681	0.6850	1.9
F	1.0837	1.4331	0.7357	2.8
P	0.5284	0.4394	1.6207	-1.6
S	0.7604	0.9021	1.2270	-0.8
T	0.6942	1.2628	1.0905	-0.7
W	1.1460	1.4842	0.6670	-0.9
Y	0.9481	1.4696	0.8108	-1.3
V	0.9126	1.8529	0.6418	4.2

Table 2: Dependence of testing accuracy on window length for each binary classifier based on coding scheme 1.

Window length	Secondary classifiers					
	H/ $\sim$ H	E/ $\sim$ E	C/ $\sim$ C	E/C	H/C	H/E
$L = 5$	0.7224	0.6828	0.6481	0.6624	0.6963	0.6815
$L = 7$	0.7435	0.6873	0.6533	0.6626	0.6594	0.6739
$L = 9$	0.7515	0.6173	0.6562	0.6628	0.6655	0.6732
$L = 11$	0.7640	0.6984	0.6597	0.6629	0.6691	0.6573
$L = 13$	0.7669	0.6915	0.6531	0.7364	0.7075	0.6903
$L = 15$	0.7679	0.6885	0.6515	0.7151	0.7007	0.6924
$L = 17$	0.7678	0.6809	0.6540	0.7229	0.6907	0.6788
$L = 19$	0.7615	0.6834	0.6484	0.7298	0.7005	0.6850
$L = 21$	0.7643	0.6784	0.6452	0.7310	0.7106	0.6721
L^*	15	15	11	13	21	15

The results are on the RS126 with SVM and coding scheme 1, with RBF kernel where $C=1.5$ and $\text{Gamma}=0.1$. Combined results of 7-fold cross-validation are shown. L^* represents the optimal window length for each binary classifier.

Table 3: Dependence of testing accuracy on window length for each binary classifier based on coding scheme 2.

Window length	Secondary classifiers					
	H/ $\sim$ H	E/ $\sim$ E	C/ $\sim$ C	E/C	H/C	H/E
$L=5$	0.7745	0.7205	0.7644	0.7284	0.6933	0.6841
$L=7$	0.7783	0.7406	0.7777	0.7301	0.7170	0.7098
$L=9$	0.7830	0.7507	0.7811	0.7313	0.7254	0.7256
$L=11$	0.7873	0.7561	0.7823	0.7323	0.7319	0.7313
$L=13$	0.7886	0.7572	0.7837	0.7354	0.7349	0.7348
$L=15$	0.7889	0.7574	0.7799	0.7364	0.7362	0.7463
$L=17$	0.7914	0.7587	0.7770	0.7345	0.7350	0.7437
$L=19$	0.7912	0.7586	0.7786	0.7291	0.7342	0.7452
$L=21$	0.7894	0.7558	0.7723	0.7281	0.7303	0.7422
L^*	17	17	13	15	15	15

Table 4: Dependence of testing accuracy on window length for each binary classifier based on coding scheme 3.

Window length	Secondary classifiers					
	H/ $\sim$ H	E/ $\sim$ E	C/ $\sim$ C	E/C	H/C	H/E
$L=5$	0.8055	0.8300	0.7722	0.8495	0.8208	0.7993
$L=7$	0.8688	0.8491	0.7759	0.8506	0.8373	0.8218
$L=9$	0.8846	0.8607	0.7774	0.8512	0.8450	0.8363
$L=11$	0.8878	0.8667	0.7819	0.8494	0.8473	0.8455
$L=13$	0.8924	0.8718	0.7802	0.8513	0.8528	0.8503
$L=15$	0.8938	0.8722	0.7893	0.8493	0.8540	0.8518
$L=17$	0.8936	0.8714	0.7777	0.8482	0.8518	0.8575
$L=19$	0.8934	0.8710	0.7748	0.8424	0.8504	0.8578
$L=21$	0.8935	0.8688	0.7723	0.8414	0.8471	0.8555
L^*	15	15	15	13	15	19

Here, $C=1.5$ $\text{Gamma}=0.1$

From Table 2, 3 and 4, we can see that the prediction accuracies of the binary classifiers are improved as the coding schemes vary from 1 to 3, which implies that structural property plays important role on determining the secondary structure prediction.

3.5 Designing the Tertiary Classifiers

The SVM used in this paper is a binary classifier, so we should design a tertiary classifier to classify H, E and C. There are many ways to design a tertiary classifier for secondary structure prediction based on binary classifier. Hua and Sun's method [14] is based on three one-versus-rest binary classifiers ($H/\sim H$, $E/\sim E$, $C/\sim C$) and three one-versus-one binary classifiers (E/C , C/H , H/E). Three cascade tertiary classifiers, SVM_TREE1 ($H/\sim H$, E/C), SVM_TREE2 ($E/\sim E$, C/H) and SVM_TREE3 ($C/\sim C$, H/E), were made up of two binary classifiers. The architecture of SVM_TREE1 is shown in Figure 2, and the architectures of SVM_TREE2 and SVM_TREE3 is similar.

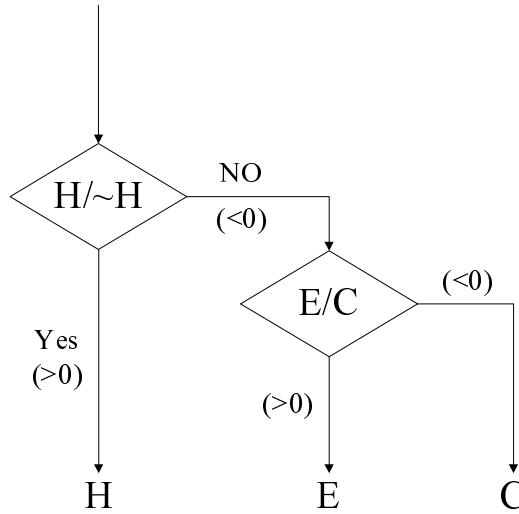


Figure 2: SVM-TREE 1. A sample will be classified as helix (H) if the output of the first binary classifier, $H/\sim H$ is larger than 0. Otherwise the second classifier, E/C will be used. If the output of E/C is larger than 0, the sample will be classified as sheet (E), otherwise it will be classified as coil(C).

Kim and Park [17] designed two additional tertiary classifiers based on a one-versus-one scheme and a DAG scheme [11]. The one-versus-one classifier for secondary structure prediction chooses the majority results based on three classifiers H/E , E/C and C/H . Many test results show that one-versus-one classifiers are more accurate than one-versus-rest classifiers because the one-versus-rest scheme often needs to deal with two data sets with very different sizes, i.e. unbalanced training data [11, 13]. However, a potential problem with the one-versus-one scheme is that the voting scheme might often suffer from incompetent classifiers. For example, while the test point is helix (H), the result from the one-versus-one classifier E/C that is not related to helix inappropriately contributes to the decision. DAG scheme can be used to avert this problem by classifying a new data point after two binary classifications. The three one-versus-one binary classifiers (E/C , H/C , H/E) construct other three tertiary classifiers called SVM_D1, SVM_D2 and SVM_D3. As an example, the architecture of SVM_D1 is shown in Figure 3.

4 Results

Mentioned in Section 3.4, the binary classifiers perform better with coding scheme 3 and window length 15 on RS126. With the same settings, we also tested on CB513 and results are shown in Table 5. From Table 5, we can see that the binary classifiers also perform very well on CB513. So we further

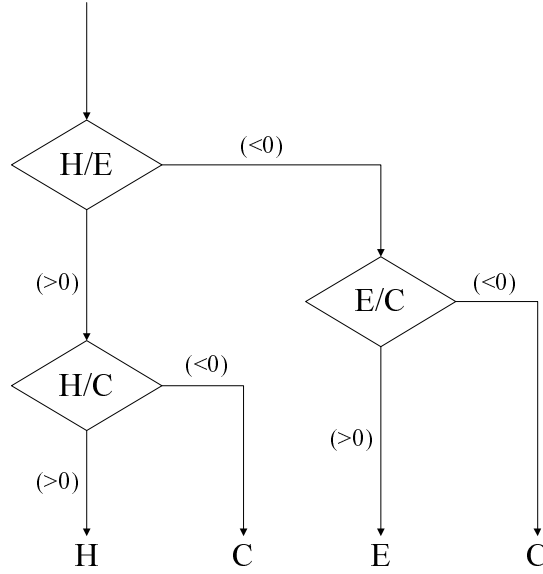


Figure 3: SVM_D1. If the testing point is predicted to be H (not E) from H/E classifier, then the H/C classifier is applied, whereas if the point is predicted to be not coil ($\sim$ H) from H/E classifier, the classifier E/C is applied to determine if it is sheet or coil.

evaluated the tertiary classifiers mentioned in Section 3.5 on CB513, still with the same settings, the results are shown in Table 6.

Table 5: The prediction accuracy of binary classifiers when $L = 15$.

	H/ $\sim$ H	E/ $\sim$ E	C/ $\sim$ C	H/E	H/C	E/C
RS126	0.8938	0.8722	0.7893	0.8518	0.8540	0.8493
CB513	0.8625	0.8718	0.7713	0.8465	0.8749	0.8074

Table 6: Accuracy Q_3 of tertiary classifiers on CB513.

	SVM_TREE1	SVM_TREE2	SVM_TREE3	SVM_D1	SVM_D2	SVM_D3
Q_3	0.7693	0.7672	0.76.93	0.7821	0.7810	0.7844
Q_H	0.7732	0.7756	0.7803	0.7860	0.7854	0.7891
Q_E	0.6714	0.6525	0.6617	0.6834	0.6879	0.6749
Q_C	0.8175	0.8204	0.8156	0.8307	0.8262	0.8378

From Table 6, we can see that SVM_D3 gets the best result of $Q_3 = 78.44\%$ on CB513, whereas Sujun Hua and Zhirong Sun got the result of $Q_3 = 73.50\%$ [14], and Hyunsoo Kim and Haesum Park got the result of $Q_3 = 76.6\%$ [17], which illustrates that, for one thing, the SVM approach is a rather good method to perform classifying problems; for another thing, the physical-chemical properties are indeed important information for protein secondary structure prediction.

From Table 6, we can also see that the results of the DAG scheme were better than those of SVM_TREE1, SVM_TREE2 or SVM_TREE3, although DAG scheme used only one-versus-one classifiers for decisions instead of all six binary classifiers. The results show that the one-versus-one scheme of DAG scheme is a good approach in the three-class classification problem, such as protein secondary structure prediction. From the architecture of this kind of scheme shown in Section 3.5, it is also easy to draw a conclusion that it can reduce the computational complexity and the difficulty of big unbalanced classification by using one-versus-one rather than one-versus-rest binary classifiers.

5 Conclusions

In this paper, we have focused on the contribution physical-chemical and structure character of amino acid to the protein secondary structure using a kernel method - support vector machine [28]. In the binary classifiers, taking more characters of amino acids into account can lead to higher accuracies. It is possible that our method can be improved by considering more physical-chemical properties of amino acids that affect the protein folding process and its structure. But when quantifying more factors, the dimension of the vector representing amino acid will become larger, which lead to the space complexity increasing.

Since the CB513 dataset was published in 1999, it does not contain the newly published protein structures. The SVM classifiers would also be improved by using larger training sets that contain new protein structures, but it also requires more memory to store data points while obtaining the optimal separating hyperplane. Furthermore, the prediction process takes longer time if the ratio between the number of support vectors and the data points becomes larger and the optimization of kernel parameters may become difficult due to the longer computing time.

Although SVMs had suffered from many difficulties, we have shown that it is a good method for protein secondary structure prediction, especially with the new designed coding scheme. It is expected that this kind of SVM is useful for the study of protein folding process.

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